Linearization and Differentials

Goals: Approximate changes using Linearizations and the language of differentials
Before Class Videos: 4.4.1

We can approximate functions by tangent lines. These are numerically easier to compute, and often good enough!

Definition:
The Linearization of $f(x)$ at the point $x = a$ is

$$\boxed{L(x) = f(a) + f'(a)(x - a)}.$$

Example 1: Estimate $\frac{1}{4.1}$ by computing the linearization at $x = 4$.

Now approximate about $x = 1$ instead:

Example 2: Use a Linearization about $x = 0$ to approximate $\frac{1}{0.1}$:

Two lessons about Linearization:

1. Approximations are typically better nearby the point where the linearization is taken.
2. The function must be differentiable at the point to even be computed.

Definition:
Interpret the differential dx as a small change in x. Then the differential $dy = f'(x)dx$ is the resulting small change in y according to the linear approximation at x.

Big Idea:

$$f(a + dx) \approx f(a) + dy.$$

Example 3: Approximate $(7.99)^{1/3}$ using the language of differentials.

Example 4: Approximate the change in area of a circle going from radius $r = 10$ to radius $r = 0.1$.

Suggested after class problems from APEX: Section 4.4: 9, 13, 17, 25, 31, 33